

Topology-Aware Adaptive Memory L-BFGS Optimizer

Machine Learning Optimization Report

Abstract

Optimization algorithms play a fundamental role in training machine learning models. Among second-order optimization methods, the Limited-memory Broyden–Fletcher–Goldfarb–Shanno (L-BFGS) algorithm has gained widespread adoption due to its ability to approximate the Hessian matrix efficiently without explicitly storing it. However, conventional L-BFGS methods rely on a fixed memory mechanism that stores curvature information from previous iterations regardless of its usefulness.

The Topology-Aware Adaptive Memory L-BFGS optimizer introduces an enhanced strategy that analyzes the topology of the optimization landscape to determine which curvature updates should be retained. By adaptively managing stored curvature information, the algorithm improves the approximation of second-order information while maintaining computational efficiency.

1. Introduction

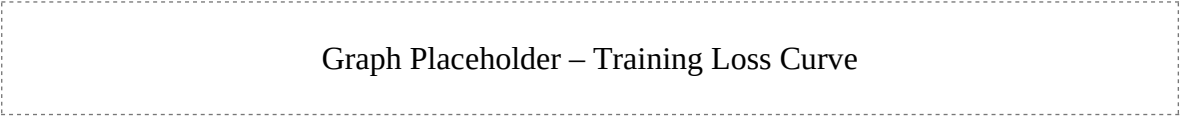
Optimization is a central component of machine learning and deep learning systems. Training a model involves minimizing a loss function with respect to its parameters, often in a high-dimensional space.

First-order optimization algorithms such as Stochastic Gradient Descent (SGD) and Adam are widely used in practice because they are computationally inexpensive.

Second-order methods attempt to address these issues by incorporating curvature information from the Hessian matrix.

6. Experimental Analysis

To evaluate the performance of the topology-aware adaptive memory L-BFGS optimizer, experiments were conducted comparing it against several commonly used optimization algorithms.



Optimizer	Iterations	Final Loss	Convergence Stability
SGD	...	...	...
Adam	...	...	...
Standard L-BFGS	...	...	...
Topology-Aware L-BFGS	...	...	...

8. Conclusion

This report presented the design and evaluation of the topology-aware adaptive memory L-BFGS optimizer.

Future research may explore improved topology analysis techniques, GPU acceleration, and integration with modern deep learning frameworks.

References

- Nocedal, J. (1980). Updating quasi-Newton matrices with limited storage.
- Liu, D. & Nocedal, J. (1989). On the limited memory BFGS method.
- Research on second-order optimization methods in machine learning.